

**Connect device for data capture**  
**Supported Devices**

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Collect new data

Collect data directly from your phone, computer, device, or development board.

Scan QR code to connect to your phone

Connect to your computer

Connect your device or development board

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Connect your device or development board

You can connect almost any device over serial using a data forwarder:

```
$ edge-impulse-data-forwarder
```

Or connect a supported device or development board:

Search supported development boards

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studio.edgeimpulse.com/studio/51070/devices

Connect your device or development board

You can connect almost any device over serial using our [data forwarder](#):

```
$ edge-impulse-data-forwarder
```

Or connect a supported [device or development board](#):

Search supported development boards

- Any device running macOS
- Arducam Pico4ML TinyML Dev Kit
- Arduino Nano 33 BLE Sense
- Arduino Nicla Vision**
- Arduino Nicla Voice

Arduino Nicla Vision

[View installation docs](#)

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Today, 11:53:14

?



Cifar10\_Image\_Classification x Arduino Nicla Vision - Edge Im x +

docs.edgeimpulse.com/docs/development-platforms/officially-supported-mcu-targets/arduino-nicla-vision

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## Installing dependencies

To set this device up in Edge Impulse, you will need to install the following software:

- Edge Impulse CLI.
- Arduino CLI.
  - Here's an [instruction video for Windows](#).
  - The [Arduino website](#) has instructions for macOS and Linux.
- On Linux:
  - GNU Screen: install for example via `sudo apt install screen`.

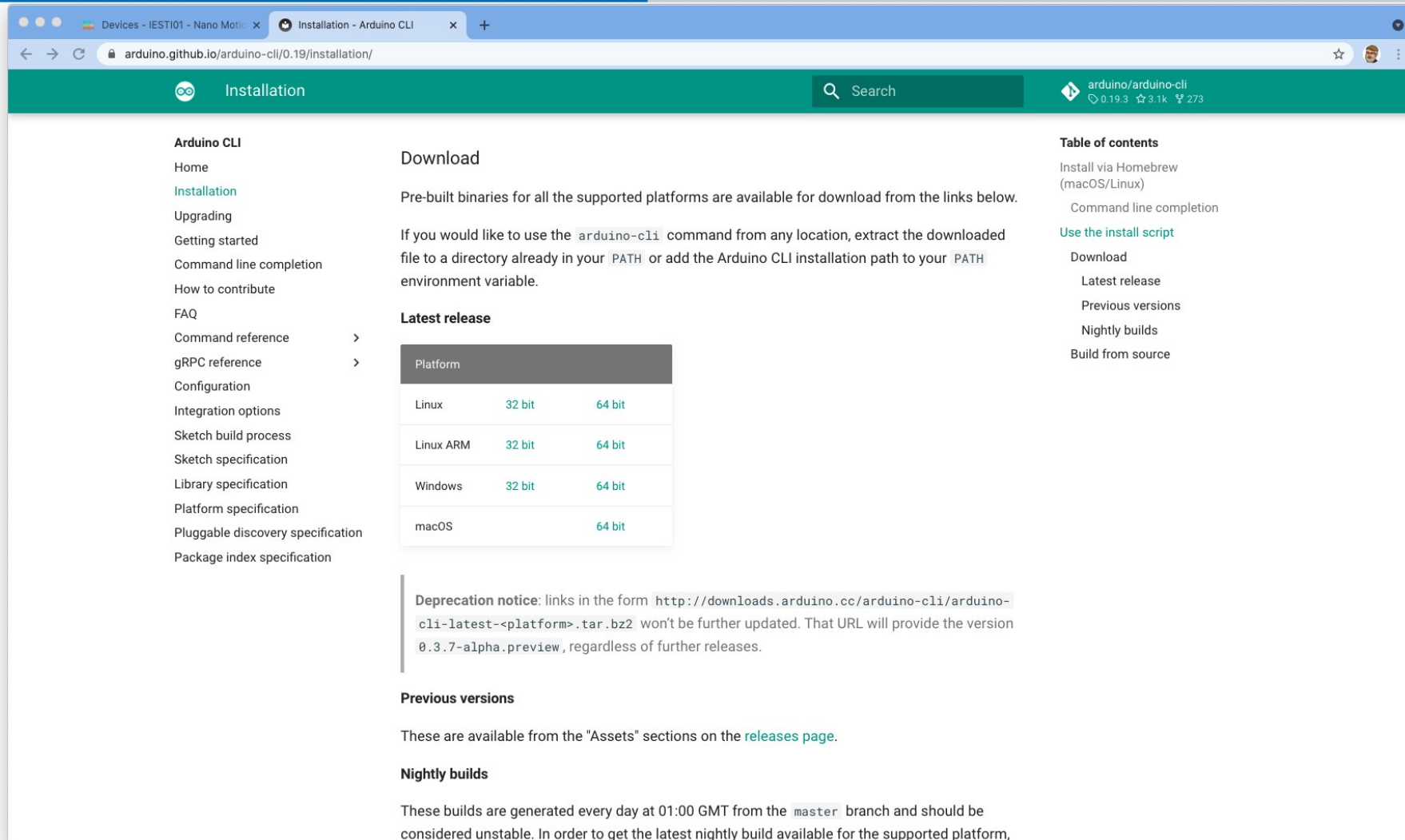
 **Problems installing the CLI?**  
See the [Installation and troubleshooting](#) guide.

## Connecting to Edge Impulse

(Note that the **1. Edge Impulse CLI** is not necessary for Arduino Nano-33. We will use **WebUSB** instead)

Go to **2. Arduino CLI**

# Arduino CLI



The screenshot shows the Arduino CLI installation page on GitHub. The page has a green header with the Arduino logo and the word "Installation". Below the header, there is a sidebar on the left with a list of links: Home, Installation (highlighted), Upgrading, Getting started, Command line completion, How to contribute, FAQ, Command reference, gRPC reference, Configuration, Integration options, Sketch build process, Sketch specification, Library specification, Platform specification, Pluggable discovery specification, and Package index specification. The main content area is titled "Download" and contains the following text: "Pre-built binaries for all the supported platforms are available for download from the links below. If you would like to use the `arduino-cli` command from any location, extract the downloaded file to a directory already in your `PATH` or add the Arduino CLI installation path to your `PATH` environment variable." Below this text is a table titled "Latest release" with the following data:

| Platform  | 32 bit                 | 64 bit                 |
|-----------|------------------------|------------------------|
| Linux     | <a href="#">32 bit</a> | <a href="#">64 bit</a> |
| Linux ARM | <a href="#">32 bit</a> | <a href="#">64 bit</a> |
| Windows   | <a href="#">32 bit</a> | <a href="#">64 bit</a> |
| macOS     |                        | <a href="#">64 bit</a> |

Below the table is a "Deprecation notice" stating: "links in the form `http://downloads.arduino.cc/arduino-cli/latest-<platform>.tar.bz2` won't be further updated. That URL will provide the version `0.3.7-alpha.preview`, regardless of further releases." Below the notice is a section titled "Previous versions" with the text: "These are available from the 'Assets' sections on the [releases page](#)." Below this is a section titled "Nightly builds" with the text: "These builds are generated every day at 01:00 GMT from the `master` branch and should be considered unstable. In order to get the latest nightly build available for the supported platform,"

On the right side of the page, there is a "Table of contents" section with the following links: "Install via Homebrew (macOS/Linux)", "Command line completion", "Use the install script", "Download", "Latest release", "Previous versions", "Nightly builds", and "Build from source".



See this video for Windows installation: <https://www.youtube.com/watch?v=1jMWsFER-Bc>

## If you're on Windows:

- Unzip the .zip file to C:\Program Files\arduino-cli
- Open System Properties > Advanced > Environment Variables
- Path under "user variables" > Edit
- Add C:\Program Files\arduino-cli

## If you're on macOS:

- I recommend using the curl method or homebrew shown on the installation page:  
<https://arduino.github.io/arduino-cli/0.21/installation/>
- if you use the curl method, it likely installs arduino-cli to ~/bin. That might not be on your path. So, you might need to run: `export PATH=$PATH:~/bin`

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Arduino Nano 33 BLE Sense

docs.edgeimpulse.com/docs/development-platforms/officially-supported-mcu-targets/arduino-nano-33-ble-sense

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### 1. Connect the development board to your computer

Use a micro-USB cable to connect the development board to your computer. Then press RESET twice to launch into the bootloader. The on-board LED should start pulsating to indicate this.



Press RESET twice quickly to launch the bootloader on the Arduino Nano 33 BLE Sense.

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### 2. Update the firmware

The development board does not come with the right firmware yet. To update the firmware:

1. [Download the latest Edge Impulse firmware](#), and unzip the file.
2. Open the flash script for your operating system ( `flash_windows.bat` , `flash_mac.command` or `flash_linux.sh` ) to flash the firmware.
3. Wait until flashing is complete, and press the RESET button once to launch the new firmware.

arduino-nano-33....zip

Show All

```
marcelo_rovai — flash_mac.command — 138x42

Last login: Tue May 23 15:47:55 on console
/Users/marcelo_rovai/Downloads/arduino-nano-33-ble-sense\ \({18}\)/flash_mac.command ; exit;
(base) marcelo_rovai@Marcelos-MacBook-Pro ~ % /Users/marcelo_rovai/Downloads/arduino-nano-33-ble-sense\ \({18}\)/flash_mac.command ; exit;
You're using an untested version of Arduino CLI, this might cause issues (found: 0.31.0, expected: 0.18.x)
Finding Arduino Mbed core...
Finding Arduino Mbed OK
Finding Arduino Nano 33 BLE...
Finding Arduino Nano 33 BLE OK
Flashing board...
Device      : nRF52840-QIAA
Version     : Arduino Bootloader (SAM-BA extended) 2.0 [Arduino:IKXYZ]
Address     : 0x0
Pages       : 256
Page Size   : 4096 bytes
Total Size  : 1024KB
Planes      : 1
Lock Regions : 0
Locked      : none
Security    : false
Erase flash

Done in 0.001 seconds
Write 352560 bytes to flash (87 pages)
[=====] 100% (87/87 pages)
Done in 13.971 seconds

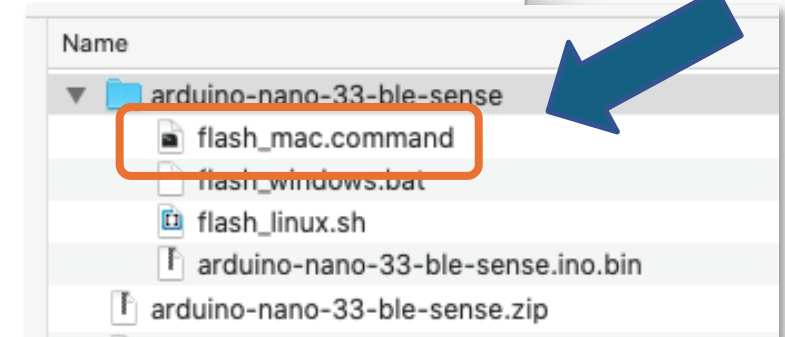
A new release of Arduino CLI is available: 0.31.0 → 0.32.2
https://arduino.github.io/arduino-cli/latest/installation/#latest-packages

Flashed your Arduino Nano 33 BLE development board.
To set up your development with Edge Impulse, run 'edge-impulse-daemon'
To run your impulse on your development board, run 'edge-impulse-run-impulse'

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

1. Press Nano-33 Reset button Twice
2. With Nano-33 LED Flashing:



3. Nano-33 LED Stop Flashing

# Windows 10

```
Prompt de Comando

Microsoft Windows [versão 10.0.19041.1052]
(c) Microsoft Corporation. Todos os direitos reservados.

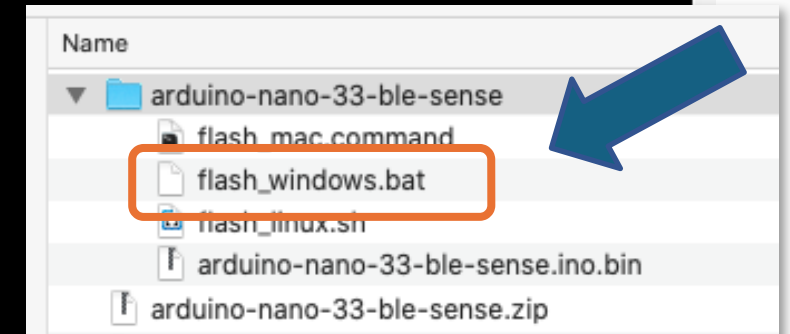
C:\Users\GUILH>arduino-cli
Arduino Command Line Interface (arduino-cli).

Usage:
  arduino-cli [command]

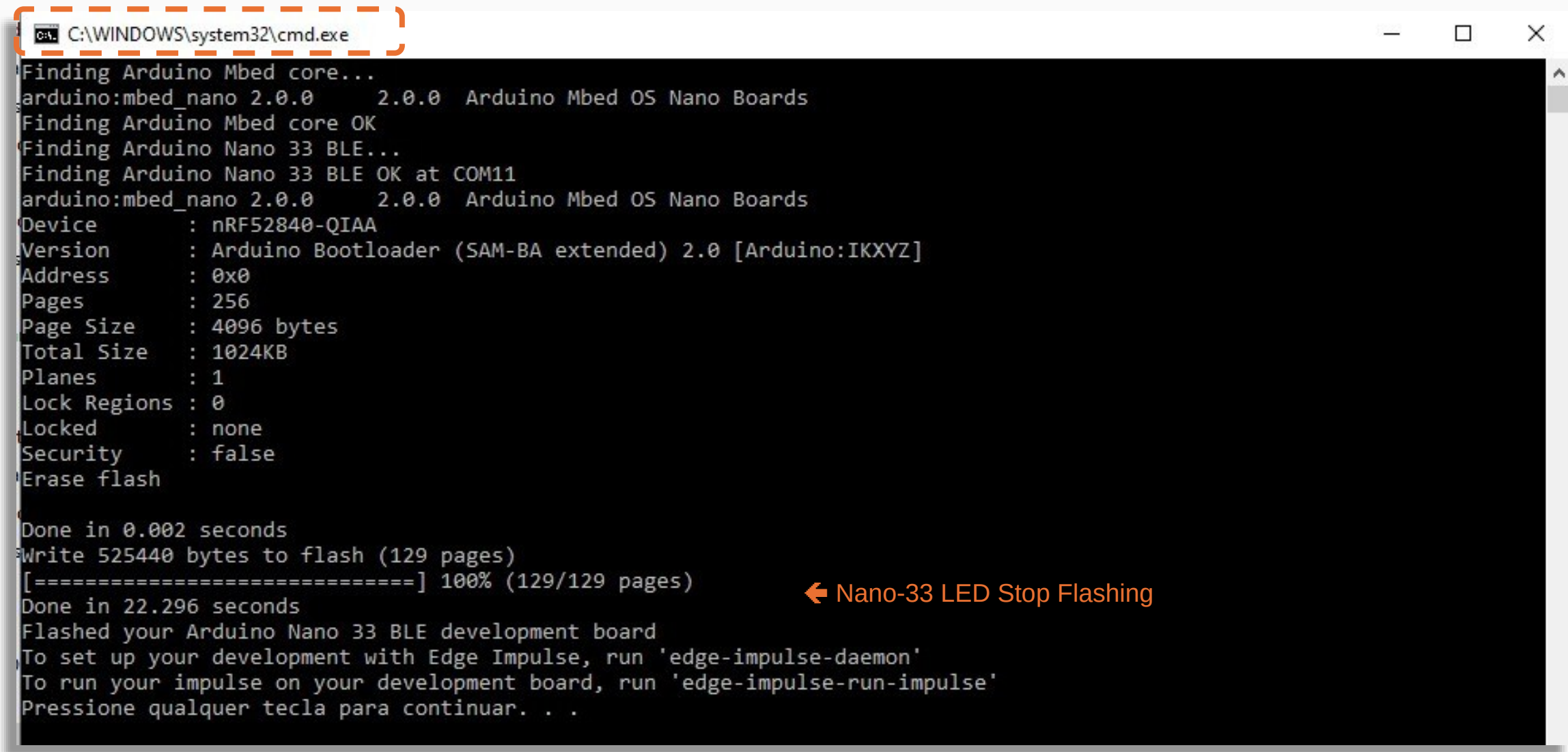
Examples:
  arduino-cli <command> [flags...]

Available Commands:
  board           Arduino board commands.
  burn-bootloader Upload the bootloader.
  cache           Arduino cache commands.
  compile         Compiles Arduino sketches.
  completion      Generates completion scripts
  config          Arduino configuration commands.
  core            Arduino core operations.
  daemon          Run as a daemon on port 50051
  debug           Debug Arduino sketches.
  help            Help about any command
  lib             Arduino commands about libraries.
  outdated        Lists cores and libraries that can be upgraded
  sketch          Arduino CLI sketch commands.
  update          Updates the index of cores and libraries
  upgrade         Upgrades installed cores and libraries.
  upload          Upload Arduino sketches.
  version         Shows version number of Arduino CLI.
```

1. Press Nano-33 Reset button Twice
2. With Nano-33 LED Flashing:



# Windows 10



```
C:\WINDOWS\system32\cmd.exe
Finding Arduino Mbed core...
arduino:mbed_nano 2.0.0      2.0.0  Arduino Mbed OS Nano Boards
Finding Arduino Mbed core OK
Finding Arduino Nano 33 BLE...
Finding Arduino Nano 33 BLE OK at COM11
arduino:mbed_nano 2.0.0      2.0.0  Arduino Mbed OS Nano Boards
Device       : nRF52840-QIAA
Version      : Arduino Bootloader (SAM-BA extended) 2.0 [Arduino:IKXYZ]
Address      : 0x0
Pages        : 256
Page Size    : 4096 bytes
Total Size   : 1024KB
Planes       : 1
Lock Regions : 0
Locked       : none
Security     : false
Erase flash

Done in 0.002 seconds
Write 525440 bytes to flash (129 pages)
[=====] 100% (129/129 pages)
Done in 22.296 seconds
Flashed your Arduino Nano 33 BLE development board
To set up your development with Edge Impulse, run 'edge-impulse-daemon'
To run your impulse on your development board, run 'edge-impulse-run-impulse'
Pressione qualquer tecla para continuar. . .
```

← Nano-33 LED Stop Flashing